

MASS-APP-014-V04 - Adaptive Intelligence, Agents & Plugin Orchestration

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1. Purpose

V04 defines the application room through which departments, approved agent profiles, plugins, NOVA, POPS, and future intelligence capabilities cooperate around authorized enterprise objectives.

V04 coordinates context, assignments, contributions, approvals, and traceability. It does not replace ENG-009 AI orchestration, ENG-006 workflow execution, the future Plugin Library, departmental authority, or executive judgment.

2. Scope

Included: agent lifecycle and application registry, department-intelligence bindings, skill routing, plugin bindings, future marketplace consumption, collaboration plans, human approval gates, adaptive-workflow recommendations, context sharing, memory utilization, Executive Cockpit integration, Dispatch integration, Design Studio integration, security, APIs, data model, and folder structure.

Excluded: model-provider orchestration, unrestricted agent autonomy, autonomous approval, direct workflow execution, plugin package certification, marketplace commerce, cross-tenant memory, department replacement, source-system mutation, and hidden chain-of-thought retention.

3. Platform Consumption Map

Dependency	Consumed capability	Delegated responsibility
APP-014 V01-V03	Sessions, memory, reasoning, decisions	Creative and decision intelligence
APP-013	Projects, assets, content, publications, visualizations	Design stewardship
ENG-003/004/005	Identity, authorization, events	Platform controls
ENG-006	Durable workflows and tasks	Workflow execution
ENG-007	Knowledge retrieval	Knowledge authority
ENG-009	Agents, prompts, tools, providers, guardrails	AI execution and orchestration
ENG-010	Assignment notifications	Notification delivery
ENG-014	Feature and environment configuration	Runtime configuration
ENG-016/026	Integration and interoperability	External connectivity
ENG-017-027	Department and enterprise intelligence	Constitutional domain authority
Future Plugin Library	Plugin packages, certification, catalog, marketplace	Plugin governance and distribution
Future Executive Cockpit	Executive presentation and command surface	Executive user experience

V04 owns application-level orchestration plans, approved participant bindings, context envelopes, contribution records, approval gates, and integration handoffs. It delegates execution and constitutional ownership.

Published events: `intelligence.plan.created`, `intelligence.assignment.requested`, `intelligence.contribution.received`, `intelligence.gate.awaiting`, `intelligence.plan.approved`, `intelligence.plan.rejected`, `intelligence.handoff.created`.

Consumed events: `agent.execution.completed`, `agent.execution.failed`, `plugin.catalog.updated`, `organizational.memory.preserved`, `decision.human.recorded`, `workflow.state.changed`, `dispatch.mission.updated`, `design.publication.created`.

4. Orchestration Model

An `IntelligencePlan` defines an objective, owner, authorized context, participant assignments, dependency graph, approval gates, output contract, and completion criteria. V04 submits each bounded assignment to its owning execution system and records signed contributions.

V04 never executes model calls, plugin code, or operational tasks directly. It coordinates contracts and state around those executions.

Lifecycle: Draft -> Validating -> Ready -> Running -> Awaiting Approval -> Approved or Rejected -> Completed -> Archived. Running may enter Paused, Failed, or Cancelled.

5. Capability Architecture

Capability	Purpose	Inputs	Outputs	Dependencies	V1 implementation	Future evolution
Agent Registry	Bind approved agent profiles to APP-014	ENG-009 profile, tenant policy, skills	AgentBinding	ENG-009, ENG-004	Tenant-approved bindings	Enterprise agent catalog
Agent Lifecycle	Govern availability and assignment	Binding, status, version	Active, suspended, retired states	ENG-009	Manual lifecycle approval	Automated health-informed suspension
Department Intelligence	Request domain-owned intelligence	Objective, domain, context	Signed domain contribution	ENG-017-027	Typed gateway requests	Broader cross-department federation
Skill Routing	Match work to approved capability	Task contract, required skills	Ranked route candidates	ENG-009, plugin catalog	Deterministic rules and explicit selection	Policy-calibrated routing
Plugin Binding	Connect approved plugins	Catalog item, permissions, configuration	Tenant PluginBinding	Future Plugin Library, ENG-004	Internal approved bindings	Certified ecosystem plugins
Marketplace Consumption	Discover compatible approved packages	Catalog and compatibility metadata	Installation request	Future Plugin Library	Architectural contract only	Governed marketplace experience
Agent Collaboration	Coordinate bounded contributions	Plan graph, assignments, evidence	Signed contribution set	ENG-009	Sequential and dependency-ordered calls	Governed parallel collaboration
Human Approval Gates	Protect accountable authority	Review package and principal	ApprovalDecision	ENG-003, ENG-004	Mandatory gates	Remains mandatory
Adaptive Workflows	Recommend plan changes from evidence	Workflow state, failures, outcomes	AdaptationProposal	ENG-006, ENG-024	Human-approved proposals	Policy-bounded dynamic adaptation
Context Sharing	Minimize and authorize shared context	Source refs, participant permissions	ContextEnvelope	V01, V02, ENG-004	Immutable bounded envelopes	Fine-grained dynamic context negotiation
Memory Utilization	Retrieve precedent and preserve outcomes	Objective, scope, verified result	Memory references and preservation handoff	V02, ENG-007	Explicit retrieval and handoff	Continuous approved learning
Executive Cockpit Handoff	Present status and approval packages	Plan summary, risks, decisions	Read model and approval request	Future Cockpit, V03	API and event contract	Integrated executive command surface
Dispatch Handoff	Exchange mission intelligence	Mission ref, context, outcome	Read-only intelligence or approved request	ENG-021	Typed handoff only	Rich field-intelligence feedback
Design Studio Handoff	Request or reference governed artifacts	APP-013 references, approved brief	Artifact request and references	APP-013	Typed handoff only	Intelligence-assisted artifact workflows

Complete architecture does not imply all future runtimes ship in V1.

6. Agent Registry and Lifecycle

AgentBinding references an ENG-009 governed agent profile and records tenant, version, approved skills, allowed tools, context policy, cost boundary, concurrency limit, approver, and lifecycle state.

States are proposed, approved, active, suspended, retired, and revoked. Only Administrator may approve or revoke. Suspension prevents new assignments but preserves history. Version changes require renewed compatibility and permission review.

The registry does not store model credentials or implement providers.

7. Skill Routing

A SkillRequirement identifies capability, input schema, output schema, sensitivity, latency class, cost class, and required assurance. SkillRoute evaluates only approved AgentBindings, PluginBindings, and department gateways.

V1 uses deterministic compatibility filters and explicit human selection. Future routing may rank candidates using performance evidence, but it cannot bypass permissions, cost limits, or approval policy.

Every routing decision stores candidates considered, exclusions, selected participant, rules applied, and selecting principal.

8. Plugin Framework and Marketplace Boundary

PluginBinding references a package governed by the future Plugin Library. It records catalog identifier, package version, compatibility range, declared capabilities, approved scopes, configuration reference, installation status, and tenant approver.

V04 defines invocation contracts and consumes catalog metadata. It does not own package storage, signing, certification, vulnerability review, marketplace commerce, licensing, installation runtime, or publisher relationships.

V1 permits only internally approved bindings. Marketplace discovery, procurement, ratings, billing, and external publishers are future Plugin Library capabilities with an established architectural home.

9. Collaboration Plans

An IntelligencePlan contains AssignmentNodes and dependency edges. Each assignment has one participant, bounded objective, authorized ContextEnvelope, output schema, timeout, retry policy, and review requirement.

V1 supports sequential and dependency-ordered execution. Parallel execution may be enabled only where assignments are independent and ENG-009 or the owning platform supports it.

ContributionRecord stores participant identity, version, input hash, output, citations, confidence, execution reference, timestamps, and error state. Contributions are advisory until accepted at a human gate.

10. Human Approval Boundaries

ApprovalGate defines the action requiring approval, eligible approver roles, quorum where required, evidence package, expiration, and downstream permission.

Agents, plugins, NOVA, POPS, and departments may recommend. They cannot approve their own contributions, change approval policy, authorize spending, communicate externally, dispatch personnel, publish artifacts, or mutate source systems.

ApprovalDecision is immutable. Correction creates an amendment. Rejection returns the plan to an explicitly authorized prior state.

11. Adaptive Workflows

V04 may create an AdaptationProposal when evidence changes, an assignment fails, risk exceeds policy, required context becomes stale, or a better approved route becomes available.

The proposal records trigger, current plan, proposed change, expected impact, risk, cost, evidence, and required approver. ENG-006 remains the workflow executor. No adaptation takes effect until required approval and a governed workflow command.

12. Context Sharing and Memory

ContextEnvelope is immutable and assignment-specific. It records source references, redactions, sensitivity, permitted uses, expiration, tenant, and authorization snapshot. Participants receive minimum necessary context.

V02 and ENG-007 provide organizational memory. V04 retrieves authorized precedent and stores only references in the plan. Verified completion outcomes may be handed to V02 for preservation after human confirmation.

Context never crosses tenants, and revocation prevents future use without deleting historical audit evidence.

13. Integration Contracts

Executive Cockpit

V04 publishes plan status, blocked gates, risks, costs, recommendations, and approval packages. The Cockpit owns executive presentation. It may submit signed approvals through V04 APIs.

Dispatch

V04 may read approved mission context and submit an intelligence handoff. Dispatch owns mission preparation, assignment, coordination, and field execution. No agent dispatches personnel.

Design Studio

V04 may reference APP-013 artifacts or submit an approved creative brief. APP-013 owns projects, components, templates, assets, publications, content, and visualizations.

14. NOVA and POPS

NOVA and POPS are governed advisory participant types executed through ENG-009.

NOVA contributes enterprise context, dependencies, patterns, and consequences. POPS contributes stewardship, constitutional alignment, continuity, and accountability guidance.

They cannot approve gates, appoint agents, install plugins, or execute operational work.

15. Security Model

- Tenant RLS applies to every V04 entity.
- Composite tenant foreign keys protect all child relationships.

- Agent and plugin credentials remain in ENG-004 governed secret references.
- ContextEnvelope access is participant-specific and time-bounded.
- Tool, cost, data, and output permissions are deny-by-default.
- Execution references are signed and auditable.
- Self-approval and contribution approval by the originating participant are prohibited.
- Revoked bindings cannot receive new assignments.
- Cross-tenant context, memory, metrics, and adaptation are prohibited.

16. Data Model

Entity	Purpose	Core integrity
IntelligencePlan	Orchestration container	Tenant lifecycle and human owner
PlanObjective	Revised objective	One active revision
AgentBinding	Approved ENG-009 profile	Version and permission approval
DepartmentBinding	Domain intelligence gateway	Constitutional owner retained
SkillDefinition	Required capability contract	Versioned schemas
SkillRoute	Auditable participant selection	Candidate and exclusion record
PluginBinding	Approved catalog binding	Package version and scopes
AssignmentNode	Bounded participant task	Plan dependency integrity
AssignmentDependency	Directed plan edge	Cycle prevention
ContextEnvelope	Immutable shared context	Sensitivity and expiration
ContextReference	Authorized source reference	Tenant and lifecycle validation
ContributionRecord	Signed participant output	Execution reference and citations
ApprovalGate	Required human checkpoint	Role and quorum policy
ApprovalDecision	Immutable human response	No self-approval
AdaptationProposal	Proposed plan change	Human approval required
OrchestrationRun	Execution attempt	Idempotency and status
InvocationRecord	Agent or plugin invocation	Input hash and output trace
MemoryUtilization	V02 memory reference	Read-only reference
IntegrationHandoff	Cockpit, Dispatch, or Design request	Idempotent destination contract
PlanOutcome	Verified result	Human verification required

17. API Inventory

Method	Route	Purpose
GET, POST	/intelligence-plans	List or create plans
GET, PATCH	/intelligence-plans/{id}	Retrieve or edit plan
POST	/intelligence-plans/{id}/validate	Validate plan
POST	/intelligence-plans/{id}/start	Start approved run
POST	/intelligence-plans/{id}/pause	Pause run
POST	/intelligence-plans/{id}/cancel	Cancel run
POST	/intelligence-plans/{id}/archive	Archive plan
GET, POST	/agent-bindings	List or propose agent bindings
PATCH	/agent-bindings/{id}	Update eligible binding
POST	/agent-bindings/{id}/approve	Approve binding
POST	/agent-bindings/{id}/suspend	Suspend binding
POST	/agent-bindings/{id}/revoke	Revoke binding

Method	Route	Purpose
GET	/department-bindings	List department gateways
GET	/skills	List skill contracts
POST	/skill-routes/evaluate	Evaluate routes
GET, POST	/plugin-bindings	List or propose plugin binding
POST	/plugin-bindings/{id}/approve	Approve binding
POST	/plugin-bindings/{id}/revoke	Revoke binding
GET, POST	/intelligence-plans/{id}/assignments	List or create assignments
POST	/intelligence-plans/{id}/assignments/{assignmentId}/invoke	Request execution
GET	/intelligence-plans/{id}/contributions	List contributions
GET, POST	/intelligence-plans/{id}/approval-gates	List or create gates
POST	/approval-gates/{id}/decisions	Record human decision
GET, POST	/intelligence-plans/{id}/adaptations	List or propose adaptations
POST	/adaptations/{id}/approve	Approve adaptation
GET, POST	/intelligence-plans/{id}/context-envelopes	List or create envelopes
POST	/intelligence-plans/{id}/memory-search	Retrieve V02 memory
POST	/intelligence-plans/{id}/nova-advisory	Request NOVA
POST	/intelligence-plans/{id}/pops-advisory	Request POPS
POST	/intelligence-plans/{id}/handoffs/executive-cockpit	Create Cockpit handoff
POST	/intelligence-plans/{id}/handoffs/dispatch	Create Dispatch handoff
POST	/intelligence-plans/{id}/handoffs/design-studio	Create Design Studio handoff
POST	/intelligence-plans/{id}/outcomes	Record verified outcome
GET	/intelligence-plans/{id}/audit	Retrieve orchestration audit

All mutations use idempotency keys. Execution requests return platform execution references, never unmanaged local processes.

18. Folder Structure and Migrations

V04 adds plans, registry, skills, routing, plugins, assignments, context, contributions, approvals, adaptations, runs, memory, and integrations. Gateways alone call ENG-006, ENG-009, V02, departments, plugins, Cockpit, Dispatch, or APP-013.

Migrations continue from 042 through 054: plans, bindings, skills, routing, assignments, context, contributions, gates, adaptations, runs, handoffs, RLS, and enforcement triggers.

19. Failure Behavior

- No compatible route: block assignment and identify missing capability.
- Participant unavailable: retry within policy or request rerouting.
- Context revoked or stale: pause affected assignments.
- Plugin or agent failure: record failure; preserve plan and prior contributions.
- Gate expires: remain awaiting approval.
- Unauthorized approval: reject without state change.
- Adaptation rejected: retain current plan.
- Integration unavailable: retain idempotent handoff for retry.

20. V1 and Future Boundary

V1 implements tenant-approved agent bindings, deterministic routing, sequential plans, immutable context envelopes, contribution records, mandatory gates, adaptation proposals, memory references, and typed integration handoffs.

Future evolution includes certified plugin marketplaces, richer department federation, approved parallel agents, health-informed routing, and policy-bounded adaptive collaboration. Their architectural homes are defined; they are not V1 runtime requirements.

21. Acceptance Verification

- Application orchestration is distinct from ENG-009 and ENG-006 execution.
- Department intelligence retains constitutional ownership.
- Plugin governance and marketplace ownership remain with the future Plugin Library.
- Human approval cannot be bypassed.
- NOVA and POPS remain advisory request types.
- Context and memory remain tenant-bound.
- Executive Cockpit, Dispatch, and Design Studio boundaries are explicit.
- Complete future intent has an architectural home without expanding V1 implementation.

22. Constitutional Boundary Statement

V04 coordinates authorized intelligence contributions. It does not own departmental intelligence, enterprise knowledge, workflows, AI providers, plugins, marketplace governance, executive decisions, field operations, or design artifacts. It cannot authorize or execute enterprise action. Constitutional authority remains with accountable humans and owning enterprise capabilities.